

Spontaneous regression of a petroclival meningioma: illustrative case

Ryuichi Noda, MD,^{1,2} Mariko Kawashima, MD, PhD,¹ Atsuya Akabane, MD, PhD,¹ and Tomohiro Inoue, MD, PhD²

¹Gamma Knife Center and ²Department of Neurosurgery, NTT Medical Center Tokyo, Shinagawa-ku, Tokyo, Japan

BACKGROUND Spontaneous regression is relatively rare in meningiomas. Since meningiomas are known to have a relationship with sex-related hormones, spontaneous regression cases related to sex hormones have been reported.

OBSERVATIONS A 66-year-old man with a petroclival meningioma was radiologically followed up for 13 years. The tumor initially measured 2.535 cm³ and 2.521 cm³ on gadolinium-enhanced magnetic resonance imaging (MRI) and fast imaging employing steady-state acquisition MRI, respectively. However, the measurements decreased in the follow-up period to 0.977 cm³ and 0.98 cm³, respectively, demonstrating spontaneous regression of an asymptomatic meningioma located in a petroclival area. Continuous meticulous long-term follow-up revealed this phenomenon.

LESSONS It is important to retain the option of conservative treatment for patients ineligible for surgery for reasons such as advanced age and coexisting diseases.

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KEYWORDS petroclival meningioma; spontaneous regression; conservative management

Since meningioma is the most common benign brain tumor, its natural history has been studied for several years¹ to verify the indications for interventional treatment, including surgical treatment, and to develop conservative treatment options, especially for asymptomatic lesions. However, previous reports have focused on the growth rate of the tumor, and only a few have described spontaneous regression, as most meningiomas grow larger or remain unchanged.^{2,3} Nevertheless, the risk factors for tumor growth have been identified;¹ thus, in patients presenting with none of these risk factors, predominantly asymptomatic cases, conservative follow-up management should be considered. Magnetic resonance imaging (MRI) has enabled the elucidation of long-term chronological changes in meningiomas, and spontaneous regression has been reported in some case reports. In this report, we present a case of spontaneous regression of a meningioma along with a literature review. We meticulously described the clinical course, measuring tumor volume using long-term MRI follow-up data.

Illustrative Case

A 66-year-old man diagnosed with a brain tumor was referred to our department for further examination. Initial MRI was performed because

he was experiencing continuous headaches but demonstrated no neurological deficits at presentation. He had no history of diabetes, hypertension, or infarction. However, he did have a preexisting gastrointestinal problem that had been detected 3 years earlier, which was later diagnosed as an indolent T-cell lymphoproliferative disorder of the gastrointestinal tract. He had no medical history of sex hormone-related medications. Gadolinium-enhanced MRI (GE-MRI) demonstrated a homogeneously enhancing brain tumor attached to the dura mater (with a positive dural tail sign) of the left petroclival area (Fig. 1A). The tumor was 25 × 8 × 25 mm in three-dimensional (3D) linear measurements. The tumor volume was obtained via 3D- assessment of GE-MRI and fast imaging employing steady-state acquisition (FIESTA) MRI using the Leksell GammaPlan software (Elekta Instrument AB), and the tumor volumes were 2.535 cm³ and 2.521 cm³, respectively. The tumor surrounded the upper half of the circumference of the basilar artery on the axial image and was partially attached to the ventral surface of the brainstem. The left trigeminal nerve was compressed by the tumor and deviated slightly laterally. The tumor was isointense on T2-weighted images, and no edema was observed on the fluid-attenuated inversion recovery sequence. The diffusion-weighted image

ABBREVIATIONS 3D = three-dimensional; ASL = arterial spin labeling; CT = computed tomography; FIESTA = fast imaging employing steady-state acquisition; GE-MRI = gadolinium-enhanced MRI; MRA = magnetic resonance angiography; MRI = magnetic resonance imaging.

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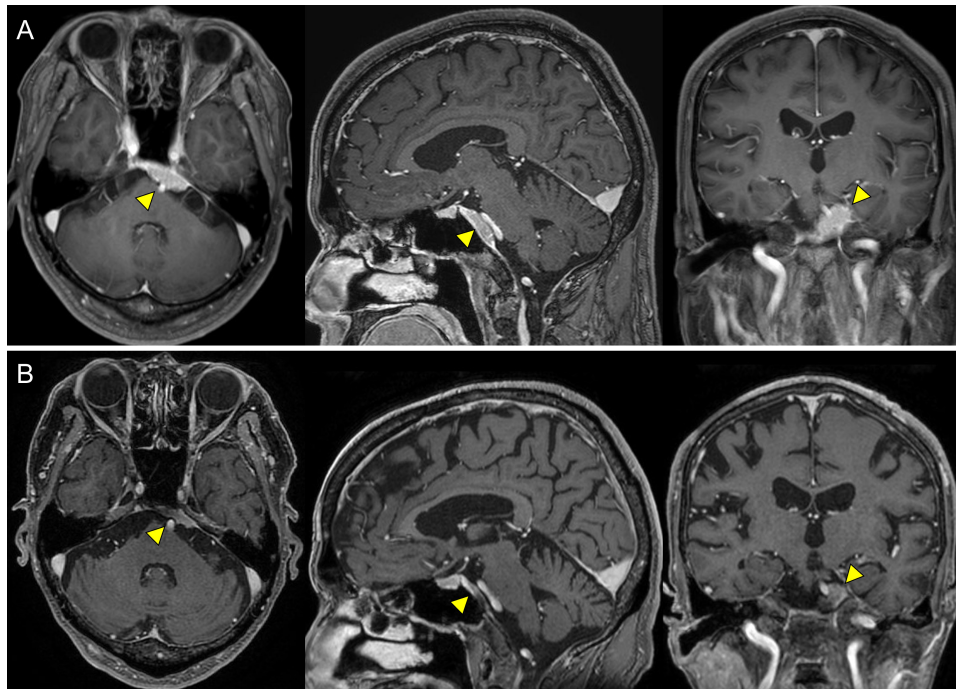


FIG. 1. Axial (*left*), sagittal (*center*), and coronal (*right*) GE-MRI. **A:** The tumor size was $25 \times 8 \times 25$ mm in 3D linear measurement and 2.535 cm^3 in volumetric measurement. The tumor is homogeneously enhanced by contrast enhancement (*arrowheads*) and has a positive dural tail sign. The tumor was surrounding the upper half of the circumference of the basilar artery and was partially attached to the ventral surface of the brainstem on the axial image. **B:** After 13 years, the tumor measured $20 \times 5 \times 20$ mm in the 3D measurement and 0.977 cm^3 in the volumetric measurement. The tumor is still homogeneously enhanced by contrast enhancement (*arrowheads*). The perivascular space between the basilar arteries is visible on the axial image, and the sagittal image showed that the tumor was not attached to the basilar artery.

showed isointensity, which indicated that the tumor was unlikely to be lymphoma, also regarding the relative stability over several years. Computed tomography (CT) showed no calcification on the tumor. The radiological diagnosis was consistent with meningioma. We refrained from performing surgical interventions, including biopsy, because of the tumor's anatomical location. Observational MRI follow-up was chosen, and the patient underwent 10 follow-up MRI procedures over the next 13 years because the tumor was asymptomatic. Most of the follow-up MRI was performed without contrast enhancement to avoid any adverse events since the initial FIESTA image demonstrated the tumor well. During the clinical course, the patient had no neurological symptoms. The tumor volume showed a gradual decrease on FIESTA during the entire follow-up period, and the tumor volumes were 0.977 cm^3 and 0.98 cm^3 on the last GE-MRI and FIESTA MRI, respectively (Figs. 1B and 2). The patient never sought treatment for the tumor. He did not receive any sex-related hormonal treatment. The patient was satisfied with the results. CT was performed incidentally 11.5 years after his first presentation, and the tumor showed an increase in calcification compared to its initial appearance (Fig. 3A). The latest MRI showed high intensity on an arterial spin labeling (ASL) image of the tumor (Fig. 3B). These findings strengthened the diagnosis of a meningioma. The latest magnetic resonance angiography (MRA) study did not reveal any atherosclerotic changes in the basilar artery, which had been initially surrounded by tumor. The latest MRI scan was also negative for any signs of ischemic brain infarction.

Patient Informed Consent

The necessary patient informed consent was obtained in this study.

Discussion

Observations

Several studies have discussed the conservative treatment of incidental meningiomas; however, most have never referred to tumor regression.^{4,5} In 2019, however, a prospective study of the natural history of incidental meningioma conducted by Behbahani et al.² showed that 4.7% of 64 patients showed volume regression. In 2022, Thomann et al.³ analyzed 240 meningiomas and reported that 2.5% of the meningiomas regressed during a mean observation period of 46.9 months. They reported that the mean age of patients with regressed meningiomas was 73.1 ± 8.5 years, with a median follow-up interval of 28.9 months and a volume regression rate of 21.4%. They identified the cause in 1 patient, which was sex-related hormones. However, apart from that patient, they did not discuss the reasons. Among all the previous studies that describe the natural history of meningiomas, these are the only 2 publications that refer to spontaneous regression. This phenomenon of spontaneous regression could be related to the elevated detection rate of asymptomatic patients brought about by current widespread diffusion and access to MRI. When an asymptomatic meningioma is detected and if the size is small and the growth rate is relatively slow, it tends to be managed by conservative MRI follow-up based on the responsible physician's experience-based decisions.

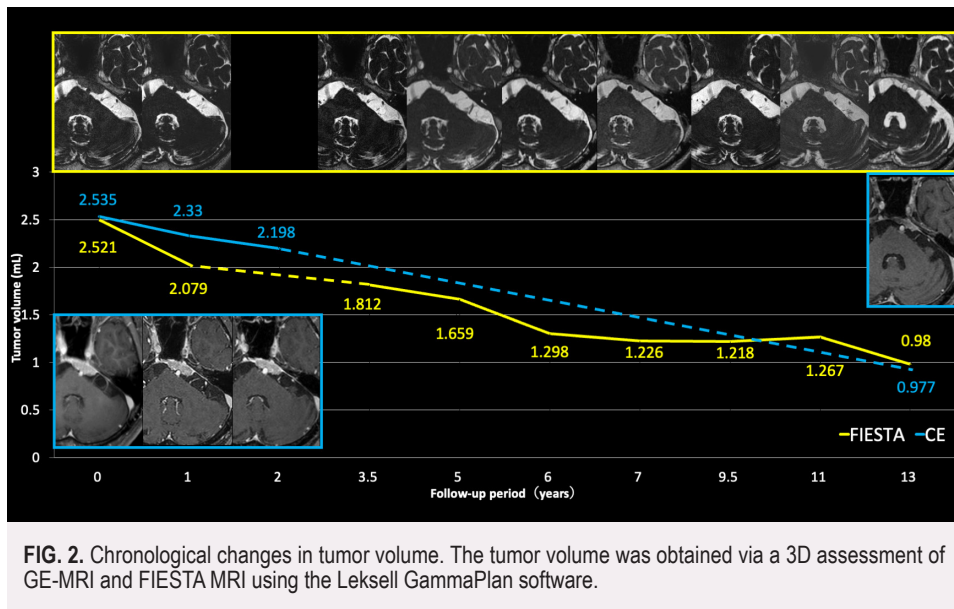


FIG. 2. Chronological changes in tumor volume. The tumor volume was obtained via a 3D assessment of GE-MRI and FIESTA MRI using the Leksell GammaPlan software.

The lack of calcification in meningioma is a known risk factor for a growing tumor,^{1,6} and careful observational follow-up is required for those with calcification. In our patient, although the initial CT showed

no calcification, we decided to perform conservative follow-up because of the absence of other risk factors (such as young age, hyperintensity on T2-weighted MRI, large size, and edema). Further multicenter studies are required to elucidate the actual frequency of spontaneous regression.

A literature review was performed by searching the PubMed database for relevant studies. The search terms used to identify articles reporting meningioma regression included “spontaneous regression” and “meningioma.” Spontaneous regression of the meningioma was defined as a decrease in the volume of the tumor for unknown reasons or caused by other medical conditions with or without any medical therapy that was not intended for treatment of the meningioma. There have been only 19 reports of 24 cases of spontaneous regression of meningiomas, including ours.^{7–24} Most of the patients were female. Sex hormone-related medical therapy and pregnancy were the most common causes of spontaneous regression. Chlormadinone acetate, norgestrel acetate, and megestrol as progesterone-related medications and cyproterone acetate as an antiandrogen agent have been reported to be the causes of spontaneous regression. Furthermore, all cases of the simultaneous regression of multiple lesions were associated with sex-related changes in hormone levels (Table 1). It is known that there is a strong connection between meningiomas and sex hormones. In general, most patients are women, and tumors are often positive for both progesterone and estrogen receptors.²⁵ A few previous cases have documented the effect of antiestrogen agents as medical therapy for meningioma.^{26,27} In our case, the patient was male and had never used any related medications.

The second cause discussed in previous publications is vascular occlusion of the feeding artery of the meningioma. Simple embolization of the feeding arteries of meningiomas contributes to tumor regression.^{28,29} In a previous study, spontaneous regression of a remnant meningioma after surgery was reported, which could be explained by mechanical occlusion of the feeding arteries due to the surgical procedure.³⁰ It is also suspected that coexisting health conditions, such as diabetes, hypertension, and hyperlipidemia, create atherosclerotic degenerative changes in the feeding artery, which causes loss of blood supply to the tumor and finally leads to

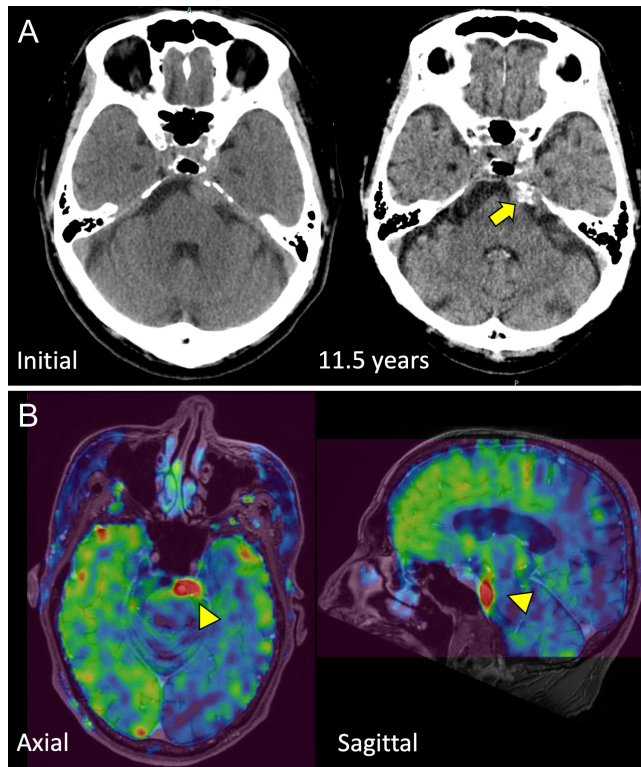


FIG. 3. A: Initial CT did not show any calcification, but CT performed 11.5 years later showed calcification of the tumor (arrow). **B:** Latest magnetic resonance image with additional sequences. ASL imaging fused with GE T1 images demonstrates high intensity, suggesting hypervascularity of the tumor (arrowheads).

TABLE 1. Literature review of cases with meningioma showing spontaneous regression

Authors & Year	Sex	Age (yrs)	Location*	Initial Size (cm ³)	Final Size (cm ³)	RR (%)	FU (mos)	Possible Cause
Pozzati et al., 2007 ⁷	F	59	Temporopolar	NA	NA	PR	24	Hormonal
			Frontal	NA	NA	CR		
			Tentorial edge	NA	NA	PR		
			Internal acoustic meatus	NA	NA	PR		
Shimizu et al., 2008 ⁸	M	80	Frontal base	φ50 mm	φ<25 mm	>50	33	Hormonal
de Almeida et al., 2009 ⁹	F	66	Parietal	5.5†	NA	90	12	Intratumoral hemorrhage
Vadivelu et al., 2010 ¹⁰	F	63	Middle fossa	4.28†	0.93†	78.3	60	Hormonal
			Supresellar	3.6†	0.9†	75.0		
			Parasagittal	0.6†	0.0	100.0		
			Medial sphenoid	0.98†	0.03†	96.9		
			Lateral greater sphenoid	49.35†	1.49†	97.0		
Gonçalves et al., 2010 ¹¹	F	46	Olfactory groove	16.0	1.9	88.0	6	Hormonal
Watanabe et al., 2012 ¹²	M	75	Parasagittal	6.1	1.1	82.0	54	Vascular
	F	72	Middle fossa	7.2	2.4	53.0	56	Vascular
	F	79	Olfactory	16.0	4.0	75.0	78	Unknown
Hirota et al., 2014 ¹³	F	66	Falx	25.5	9.9	61.2	84	Vascular
Kerschbaumer et al., 2016 ¹⁴	F	32	Sphenoid	53.1	27.6	48.0	2	Hormonal
Witherick et al., 2016 ¹⁵	F	75	Falx	NA	NA	NA	84	Unknown
Hoegestoele & Berg-Johnsen, 2016 ¹⁶	M	59	Sphenoid	29.9	6.7	77.6	37	Alpha-1 adrenoceptor agonist
Galloway et al., 2017 ¹⁷	F	56	Falx	NA	NA	NA	120	Interferon beta
Kalamarides & Peyre., 2017 ¹⁸	F	26	Parietal	68.0	13.0	81.0	12	Hormonal
			Parasagittal	1.6	0.3	80.0		
	F	43	Temporal	64.0	11.0	83.0	12	Hormonal
			Posterior petrous bone	6.4	5.3	17.0		
			Optic nerve	2.3	1.0	53.0		
Passeri et al., 2019 ¹⁹	F	37	Clinoid	54.5	13.1	76.0	12	Hormonal
	F	68	Clivus	12.4	10.3	16.9	4	Hormonal
	F	54	Frontal	5.7	5.2	9.0	7	Hormonal
			Clinoid	9.4	7.9	16.0		
Kumaria et al., 2020 ²⁰	F	58	Sphenoid	36.0	11.2	68.9	84	Unknown
Shahin et al., 2021 ²¹	F	40	Falx	φ11 mm	φ8 mm	27.3	4	Hormonal
			Parietal	φ10 mm	φ5 mm	50.0		
			Frontal	φ11 mm	φ7 mm	36.4		
Takada et al., 2022 ²²	F	66	Posterior fossa	2.3	0.5	78.3	144	Unknown
Kim et al., 2022 ²³	F	73	Sphenoid	58.6	37.2	36.6	120	Vascular
Bailey et al., 2024 ²⁴	F	NA	Sphenoid	14.9	3.2	78.5	60	Hormonal
			Planum sphenoidale	1.9	0.4	78.9		
			Sphenoid wing	0.3	0.1	66.7		
Present case	M	66	Petroclival	2.5	1.0	62.5	154	Unknown

CR = complete response; FU = follow-up; NA = not available; PR = partial resolution; RR = regression rate; φ = diameter.

Median RR = 75.0% (9.0%–100%); median initial size = 7.2 (0.3–68.0) mL; median FU = 45.5 (2.0–154.0) months.

* The locations are described as they were in the original article.

† The volume was calculated by $(a \times b \times c)/2$.

its regression.^{12,13} However, confirming atherosclerotic changes in a feeding artery is almost impossible because the vessel itself is usually too thin to be inspected on radiological imaging. In our case, the latest CT performed after 11.5 years showed intratumoral calcification that had not existed on the initial CT. This phenomenon was also

observed in 1 case report¹³ in which vascular reasons were possibly the cause of spontaneous regression. This phenomenon resembles the tumor growth pattern change with the progression of calcification demonstrated by Nakasu et al.³¹ However, the proposed mechanisms are largely correlative.

In our case, the patient had no medical history of atherosclerosis. Furthermore, the ASL image at the latest MRI showed that the tumor was still hypervascular, and MRA did not reveal any atherosclerotic changes in the various vessels of the vertebrobasilar artery. Although the mechanism of regression in our case is still not certain, another possible etiology is that the growing tumor invasion could have caused mechanical compression and occlusion, leading to the disruption of the smallest feeding arteries that nourish the tumor. The initial tumor was attached to the surrounding basilar artery. Some basilar perforating arteries may have been mechanically affected by tumor invasion. It is important to accumulate case reports of the spontaneous regression of meningioma not only because it is rare but also because it could shed light on the potential mechanisms and become a clue to novel medication therapy for meningioma.

Recent advances in molecular biology studies of meningioma are starting to elucidate the clinical features of the tumor.³² As regressive meningioma specimens are rarely obtained, molecular biological investigations have not been conducted. However, the spatial phenotype may be useful because it may signify a genotype.³³ According to previous reports, most regressed lesions are located at the antero-medial cranial base or convexity, whereas those in the posterior fossa are relatively rare. Nevertheless, further accumulation of cases is required.

Lessons

We encountered a rare case of the spontaneous regression of an asymptomatic meningioma located within a petroclival area. We were able to depict the patient's medical course by continuing a meticulous long-term follow-up. It is important to retain the option of conservative treatment for patients who are not eligible for surgery for specific reasons, such as advanced age and coexisting diseases.

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Disclosures

The authors report no conflict of interest concerning the materials or methods used in this study or the findings specified in this paper.

Author Contributions

Conception and design: Noda, Inoue. Acquisition of data: Noda. Analysis and interpretation of data: Noda. Drafting the article: Noda. Critically revising the article: Kawashima, Inoue. Reviewed submitted version of manuscript: Noda. Approved the final version of the manuscript on behalf of all authors: Noda. Study supervision: Akabane, Inoue.

Correspondence

Ryuichi Noda: NTT Medical Center Tokyo, Shinagawa-ku, Tokyo, Japan. rnm46_8447@yahoo.co.jp.